



Solve with us 9 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1Key Points: 1

- **Scalar-valued multivariable functions:** A scalar-valued multivariable function is a function $f: D \rightarrow \mathbb{R}$, where D is a domain in $\mathbb{R}^n$ where $n > 1$.
- **Vector-valued multivariable functions:** A vector-valued multivariable function is a function $f: D \rightarrow \mathbb{R}^m$, where D is a domain in $\mathbb{R}^n$ where $n, m > 1$.

1 point

Which of the following functions are scalar-valued multivariable functions?

☐

$$f(x, y) = xy + \sin(xy)$$

☐

$$f(x, y, z) = \ln(x + y + z)$$

☐

$$f(x, y, z) = (x^2, z, y + xz)$$

☐

$$f(x, y) = (xy, 0)$$

☐

$$f(x, y) = xy$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y) = xy + \sin(xy)$$

$$f(x, y, z) = \ln(x + y + z)$$

$$f(x, y) = xy$$

1 point

Which of the following functions are vector-valued multivariable functions?

☐

$$f(x, y) = xy + \sin(xy)$$

☐

$$f(x, y, z) = \ln(x + y + z)$$

☐

$$f(x, y, z) = (x^2, z, y + xz)$$

☐

$$f(x, y) = (xy, 0)$$

☐

$$f(x, y) = xy$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y, z) = (x^2, z, y + xz)$$

$$f(x, y) = (xy, 0)$$



Key Points: 2Key Points: 2

- Composition of functions:Composition of functions: Let $D \subseteq \mathbb{R}^n$ $D \subseteq \mathbb{R}^n$ and $f: D \rightarrow \mathbb{R}^m$ $f: D \rightarrow \mathbb{R}^m$ be a multivariable function. Let $g: E \rightarrow \mathbb{R}^p$ $g: E \rightarrow \mathbb{R}^p$ be a function on E where $\text{Range}(f) \subseteq E \subseteq \mathbb{R}^m$ $\text{Range}(f) \subseteq E \subseteq \mathbb{R}^m$.

We define a multivariable function $g \circ f: D \rightarrow \mathbb{R}^p$ $g \circ f: D \rightarrow \mathbb{R}^p$ called the composition of f and g defined as $g \circ f(x) = g(f(x))$ $g \circ f(x) = g(f(x))$, for all $x \in D$ $x \in D$.

1 point

If two functions $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ $g: \mathbb{R} \rightarrow \mathbb{R}$ are defined as $f(x, y) = (x - y)$ $f(x, y) = (x - y)$ and $g(x) = \sin x$ $g(x) = \sin x$, then which of the following functions represents $g \circ f$ $g \circ f$?

☐

$\sin x - \sin y$ $\sin x - \sin y$.

☐

$\sin(x - y)$ $\sin(x - y)$.

☐

$x \sin x - y \sin y$ $x \sin x - y \sin y$.

☐

$\sin x y \sin xy$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sin(x - y)$ $\sin(x - y)$.

1 point

If two functions $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}^2$ $g: \mathbb{R} \rightarrow \mathbb{R}^2$ are defined as $f(x, y) = xy$ $f(x, y) = xy$ and $g(t) = (e^t, e^{-t})$ $g(t) = (e^t, e^{-t})$, then which of the following functions represents $g \circ f$ $g \circ f$?

☐

$g \circ f(x, y) = e^{xy}$ $g \circ f(x, y) = e^{xy}$.

☐

$g \circ f(x, y) = 1$ $g \circ f(x, y) = 1$.

☐

$g \circ f(x, y) = (e^{xy}, e^{-xy})$ $g \circ f(x, y) = (e^{xy}, e^{-xy})$.

☐

$g \circ f(x, y) = (e^x, e^{-x})$ $g \circ f(x, y) = (e^x, e^{-x})$.

☐

$g \circ f(x, y) = (1, 1)$ $g \circ f(x, y) = (1, 1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$g \circ f(x, y) = (e^{xy}, e^{-xy})$ $g \circ f(x, y) = (e^{xy}, e^{-xy})$.

Key Points: 3Key Points: 3

- Partial derivatives: Partial derivatives: Let $f(x_1, x_2, \dots, x_n)$ be a function defined on a domain D in $\mathbb{R}^n$. The partial derivative of f with respect to x_i is the function denoted by $f_{x_i}(x)$ or $\frac{\partial f}{\partial x_i}(x)$ and defined as

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \rightarrow 0} \frac{f(x + he_i) - f(x)}{h}$$

where e_i is the unit vector $(0, 0, \dots, 0, 1, 0, \dots, 0)$ (at i -th coordinate there is 1, and 0 at all the other coordinates).

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f((x_1, x_2, \dots, x_n) + h(0, 0, \dots, 0, 1, 0, \dots, 0)) - f(x_1, x_2, \dots, x_n)}{h}$$

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{h}$$

- To calculate the partial derivative with respect to x_i , think of f only as a function of x_i while treating all other variables as constants. Then calculate it as the derivative of a function of one variable.
 1 point
 Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(x, y) = x^2 + y^2$. Which of the following function will represent f_x or $\frac{\partial f}{\partial x}$?

☐

$2x + 2y$.

☐

$2x$.

☐

$2x + y^2$.

☐

$x + y^2$.

No, the answer is incorrect.

Score: 0

Feedback:

Treat y as constant.

Accepted Answers:

$2x$.

1 point

Consider the function $f: D \rightarrow \mathbb{R}$ where $D \subset \mathbb{R}^2$, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$(0, 0)$

What will be the partial derivative of f with respect to x at the point $(0, 0)$?

☐

11.

☐

00.

☐

22

☐ Cannot be determined.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

00.

Key Points: 4Key Points: 4

- Directional derivative: Directional derivative: The rate of change of $f(x_1, x_2, \dots, x_n)$ in the direction of the unit vector $u = (u_1, u_2, \dots, u_n)$ is called the directional derivative and is denoted by $D_u f(x_1, x_2, \dots, x_n)$. The definition of the directional derivative is:

$$D_u f(x_1, x_2, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1 + u_1 h, x_2 + u_2 h, \dots, x_n + u_n h) - f(x_1, x_2, \dots, x_n)}{h}$$

- To calculate the directional derivative of a function f in a particular direction, we have to first find the unit vector u in that direction and then compute $D_u f$.
- The partial derivatives are special cases of the directional derivative where $u = e_i$ where e_i is the i -th vector of the standard ordered basis of R^n .

1 point

Consider the function $f: R^2 \rightarrow R$, such that $f(x, y) = x + y$. Which of the following options will denote the directional derivative of f in the direction of the vector $v = (1, 1)$?

[Hint: The given vector v is not a unit vector. Convert v to a unit vector $u = \frac{v}{\|v\|}$.]

☐ 2

☐

$x + y$

☐

$\sqrt{2}$

☐

$\sqrt{2}(x + y)$

☐

$(x + y)$

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

$\sqrt{2}$

☐

Key Points: 5Key Points: 5

- Limit of a scalar valued multivariable function: Limit of a scalar valued multivariable function: If there exists some real number L such that $f(x_1, x_2, \dots, x_n)$ approaches L as $(x_1, x_2, \dots, x_n)$ approaches $(a_1, a_2, \dots, a_n)$, then we write

$$\lim_{(x_1, x_2, \dots, x_n) \rightarrow (a_1, a_2, \dots, a_n)} f(x_1, x_2, \dots, x_n) = L$$
- For functions of one variable $f(x)$, there are only two ways in which x can approach a point a , from left or from right. But for multivariable functions, $(a_1, a_2, \dots, a_n)$ can be approached along any curve.
- If C is a curve passing through $(a_1, a_2, \dots, a_n)$, the limit of f at $(a_1, a_2, \dots, a_n)$ along the curve C exists and equals L if $f(x_1, x_2, \dots, x_n)$ approaches L as $(x_1, x_2, \dots, x_n)$ approaches $(a_1, a_2, \dots, a_n)$ where $(x_1, x_2, \dots, x_n)$ belongs to C .
- The limit of the function at $(a_1, a_2, \dots, a_n)$ along the curve C equals L for each curve C passing through $(a_1, a_2, \dots, a_n)$ precisely when the limit of f at $(a_1, a_2, \dots, a_n)$ equals L .
- To show that the limit of f at $(a_1, a_2, \dots, a_n)$ does not exist, one should either produce a curve C so that the limit of f at $(a_1, a_2, \dots, a_n)$ along C does not exist or produce two curves so that the limits of f at $(a_1, a_2, \dots, a_n)$ along both curves are not equal.
- Continuity of a function: Continuity of a function: If limit of a function $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$, exists at a point $(a_1, a_2, \dots, a_n)$, and the functional value and the limiting value are the same, i.e.,

$$\lim_{(x_1, x_2, \dots, x_n) \rightarrow (a_1, a_2, \dots, a_n)} f(x_1, x_2, \dots, x_n) = f(a_1, a_2, \dots, a_n)$$

then we say f is continuous at the point $(a_1, a_2, \dots, a_n)$.

Consider the function $f: D \rightarrow \mathbb{R}$ where $D \subset \mathbb{R}^2$, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Answer the following question 8 to 13 using the above information.

If f approaches L as (x, y) approaches the origin along the X -axis, then find the value of L .

No, the answer is incorrect.

Score: 0

Feedback:

Along the X -axis, $y = 0$.

Accepted Answers:

(Type: Numeric) 0

1 point

If f approaches L as (x, y) approaches to the origin along the Y -axis, then find the value of L .

No, the answer is incorrect.

Score: 0

Feedback:

Along the Y -axis, $x = 0$.

Accepted Answers:

(Type: Numeric) 0

1 point

If f approaches L as (x, y) approaches to the origin along the straight line $y = x$, then find the value of L .

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

(Type: Numeric) 0.5

1 point

1 point

If f approaches L as (x, y) approaches to the origin along the straight line $y = 2x$, then find the value of L .

☐

0.

☐
 $\frac{1}{2}$.
☐
 $\frac{2}{9}$.
☐

11.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

 $\frac{2}{9}$.

1 point

If f approaches L as (x, y) approaches to the origin along the straight line $y = mx$, then find the value of L .

☐

0.

☐
 $\frac{1}{m^2}$.
☐
 $\frac{m}{1+m^3}$.
☐

11.

No, the answer is incorrect.

Score: 0

Feedback:**Accepted Answers:**

$$\frac{m}{1+m^3}1+m3m.$$

1 point

Choose the set of correct options.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.
☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$
☐

f is not continuous at the origin.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{1}{2}$
No, the answer is incorrect.**Score: 0****Accepted Answers:**
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

f is not continuous at the origin.

1 point

Consider the function $f: D \rightarrow \mathbb{R}$ where $D \subset \mathbb{R}^2$, defined as:

$$f(x,y) = \begin{cases} \frac{y}{x^2+y} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Choose the set of correct options.

[Hint: Approach the origin using the curves $y = x^2$ and $y = 2x^2$.]☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.
☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$
☐

f is not continuous at the origin.

☐
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 1$
No, the answer is incorrect.**Score: 0****Accepted Answers:**
 $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

f is not continuous at the origin.

Key Points: 6Key Points: 6

- Gradient of a function: Gradient of a function: If f is a scalar valued function defined on a domain D in R^n , then the gradient ∇f is defined as the vector valued function $(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n})$.
- If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \dots, a_n)$, then the directional derivative of the function f at $(a_1, a_2, \dots, a_n)$ in the direction of a vector $\mathbf{v}$ is given by the dot product of the gradient of the function at $(a_1, a_2, \dots, a_n)$ with the unit vector in the direction of $\mathbf{v}$ i.e.,

$$D_{\mathbf{v}} f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n) \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad D_{\mathbf{v}} f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n) \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

1 point

Find the directional derivative of the function $f: R^2 \rightarrow R$ such that $f(x, y) = 3x^2 - 2xy$ in the direction of the vector $(3, 4)$.

☐

$$2x - \frac{6y}{5} 2x - 56y.$$

☐

$$10x - 6y 10x - 6y.$$

☐

$$10x 10x.$$

☐

$$2x 2x.$$

☐

$$-\frac{6y}{5} - 56y.$$

No, the answer is incorrect.**Score: 0****Accepted Answers:**

$$2x - \frac{6y}{5} 2x - 56y.$$

1 point

Find the directional derivative of the function $f: R^2 \rightarrow R$ such that $f(x, y) = 3x^2 + 2y^2$ at the point $(1, 1)$ along the direction of its gradient vector $\nabla f(1, 1)$.

[Hint: First find out the unit vector in the direction of the vector $\nabla f(1, 1)$.]

☐

$$22$$

☐

$$\sqrt{13} \quad 13$$

☐

$$2\sqrt{13} \quad 13$$

☐

$$2\sqrt{2} \quad 2$$

☐

$$\sqrt{2} \quad 2$$

No, the answer is incorrect.**Score: 0**

Accepted Answers:

$2\sqrt{132}$ 13

$\sqrt{}$